Peer-review

Title of the paper: A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

What are the strengths of the paper? What are the contributions?

The paper presents a robust, innovative prototype for low-cost motion tracking by integrating IMU signals with video-based pose estimation in a standard web browser, addressing key barriers like expense and hardware dependency in healthcare and rehabilitation applications. Its strengths lie in the fully client-side implementation using TensorFlow.js and Chart.js, which ensures accessibility without installations or servers, while incorporating practical features such as event-based synchronization, session export/import via ZIP archives, and annotation tools for reproducible analysis. The development approach demonstrates methodological rigor through iterative prototyping, performance optimizations like frame skipping and confidence thresholding, and clear handling of ethical concerns including local data processing and informed consent disclaimers.

The primary contributions include a functional browser application that enables side-by-side visualization of 2D keypoints from MoveNet (Thunder for uploads, Lightning for live) overlaid on video with synchronized IMU channels (accelerometer, gyroscope, magnetometer) derived from Movesense CSV files at 104 Hz. It advances low-complexity motion analysis by introducing manual event-based temporal alignment via a constant offset, supporting annotation of events like claps or jumps with labels and notes, and generating TXT reports for documentation. Additionally, the work contributes a user-friendly workflow for exploratory inspection in non-lab settings, with appendices detailing development history and usage guides, alongside forward-looking suggestions for sensor fusion and automated synchronization.

How is the presentation of the paper? Well-structured, good language, easy to follow, and at right level for the intended audience?

The report is clearly presented and follows a logical academic structure, progressing from motivation and background to methods, results, discussion and conclusion. This clear organization makes the paper easy to understand how the different components of the system relate to one another. The language is generally appropriate for a technical academic audience. The methods section is detailed with clear explanations of design choices, implementation decisions and limitations. The step-by-step descriptions of the system workflow, synchronization procedure and user interaction contribute to the paper being easy to follow. The inclusion of figures, appendices, and a user guide further improves clarity, especially for readers interested in reproducing the work.

Does the paper make a good use of references? Too few? Too many? The right ones?

The report provides only one reference highlighting the speed-quality trade off the MoveNet pose detector as provided by the TensorFlow team. Besides this, hyperlinks are provided to aid understanding of IMU sensors as well as the specific sensor used by the project team. While these sources are appropriate, they could be referenced in the References sections as per

standard practices. The report, however, is grossly under-referenced and could use a lot more citations, especially from journal articles.

Does the paper make a good use of figures, tables, and formulas? Too few? Too many? The right ones?

The paper makes a generally effective use of visual aids to illustrate the system's architecture, though it lacks tabular data to summarize the findings.

The use of figures is strategic and helpful. Specifically, the flowcharts in Figures 2, 3, and 4 are excellent; they clearly visualize the logic behind data export and report generation, effectively replacing what would otherwise be dense textual descriptions. Additionally, the comprehensive set of screenshots in Appendix B serves its purpose well as a visual user guide, documenting the entire workflow from video upload to annotation. Regarding formulas, their usage is minimal but appropriate for the project's scope; the synchronization offset equation (Δt) adequately defines the core technical concept without unnecessary complexity.

However, the paper notably lacks tables, which hinders the readability of the 'Methods' and 'Results' sections. A comparison table detailing the specifications and trade-offs between the MoveNet Lightning and Thunder models (discussed in section 2.2) is essential and currently missing. Similarly, the 'Results' section relies entirely on text; a table summarizing the test cases (e.g., video duration, movement types, and sync offsets) would provide a much clearer overview of the system's performance. Finally, while the content of the figures is good, the visual quality of the screenshots (particularly Figure 1) is too dark, making UI elements difficult to distinguish.

How does the paper deal with its chosen topic/problem/question? Do you agree with its approach and conclusions? How can it be improved?

The implementation demonstrates good technical decision making. Using TensorFlow.js for browser-based pose estimation removes installation barriers, and the choice of different MoveNet models for different contexts (Thunder for uploads, Lightning for live) shows thoughtful consideration of the speed vs accuracy trade-off. The iterative development approach, building and validating IMU and video modules separately before integration, was sensible.

The frame skipping and playback rate adjustments are reasonable optimizations for running on consumer hardware, and the paper is transparent about these compromises. The session management system with localStorage and ZIP export is well designed for practical use.

The manual synchronization implementation works as a proof of concept and is appropriately simple for an initial prototype. Given the project scope, having users mark reference events is a pragmatic starting point. The constant offset assumption is reasonable for short recordings, which seems to be the primary use case here.

The main area needing strengthening is validation of the synchronization accuracy. Even simple measurements would help. For example, adding some quantification of synchronization accuracy, such as having 2 or 3 users mark the same clap event in the same video and IMU pair and reporting the variance in their time offsets. This would not require major additional work

but would give readers confidence in the method's reliability. Similarly, verifying that the IMU actually samples consistently at 104 Hz, perhaps by comparing the reconstructed timeline with known event intervals, would validate a key assumption.

The paper mentions that "performance variability across devices" is a limitation but providing basic performance measurements on 2 or 3 different devices, like pose estimation frame rates on a typical laptop vs a lower end device, would be useful.

The annotation feature is nicely implemented, though the event classification (Jump, Clap, Other) currently seems underutilized. Perhaps noting in future work that these classifications could enable automated analysis would tie it together. The core implementation is solid for a course project and successfully demonstrates the feasibility of the approach.